Chapter 10 Answers

Problem 1

Suppose that B, M and R are present. By Problem 1 of Chapter 9, $M(BxM)$ is a fixed point of every element x. Now, $BxM = RMBx$. Thus $M(BxM) = M(RMBx) = BM(RMB)x$, which is to say that $BM(RMB)x = M(BxM) = xM(BxM) = x(BM(RMB)x)$, thus showing that $BM(RMB)$ is a sage.

Problem 2

In a similar manner as in Problem 1, $BM(CBM)$ is a sage, as $BM(CBM)x = M(CBMx) = M(BxM)$ and thus $BM(CBM)x = M(BxM) = xM(BxM) = x(BM(CBM)x)$.

Problem 3

Recall Problem 30 of Chapter 8, which states that, if $\mathcal{C}$ contains a lark L and a mocker M, and if the composition condition holds, then $\mathcal{C}$ contains a sage, in particular, any element θ which combines M with L is a sage. In addition, recall that, by Problem 1 of Chapter 9, if B and M are present, the composition condition holds, in particular, for any elements x and y, Bxy combines x with y. Thus, if we assume the combinators B, M and L, the composition conditions holds, and any element θ which combines M with L is a sage. In particular, the element BML is a sage. Now, if the combinators B, M and R were present, we could, by Problem 35 of Chapter 9, take RMB to be L, meaning $BM(RMB)$ is a sage. In a similar fashion, if the combinators B, M and C were present, then, by the same problem, we could take CBM to be L, meaning $BM(CBM)$ is a sage.

Problem 4

Suppose the combinators B, M and W are present. By Problem 35 of Chapter 9, we could take L to be BWB, meaning, by Problem 3, $BM(BWB)$ is a sage.

Problem 5

(Smullyan seems to assume W is also present for this problem, as I do not think that W is derivable from B, L and Q alone) Recall that $Lx(Lx)$ is a fixed point for every x. Take the sage to be $W(QL(QL))$,

as $Lx(Lx) = QL(Lx)x = QL(QL)xx = W(QL(QL))x$, meaning $W(QL(QL))x = Lx(Lx) = x(Lx(Lx)) = x(W(QL(QL))x)$.

Problem 6

Recall that, because $CBxyz = Byxz = y(xz)$, CB is a queer bird. In addition, recall that, by Problem 35 of Chapter 9, BWB is a lark, as $BWBxy = W(Bx)y = Bxyy = x(yy)$. Thus, by Problem 5, $W(QL(QL))$, that is, $W(CB(BWB)(CB(BWB)) = W(B(CB(BWB))(BWB))$ is a sage.

Problem 7

Suppose the combinators Q, M and L are present. Recall that $Lx(Lx)$ is a fixed point of every x. Now, $Lx(Lx) = M(Lx) = QLMx$. Thus take the sage θ to be QLM, as $x(QLMx) = x(Lx(Lx)) = Lx(Lx) = QLMx$.

Problem 8

Suppose the combinators Q and M are present. Take L to be QM, as $QMxy = x(My) = x(yy)$, meaning it is a lark. Thus, by Problem 7, QLM, that is, $Q(QM)M$, is a sage.

Problem 9

Suppose the combinators L and S are present. Recall that $Lx(Lx)$ is a fixed point for every x. Take the sage to be SLL, as $Lx(Lx) = SLLx$, meaning $x(SLLx) = x(Lx(Lx)) = Lx(Lx) = Lx(Lx) = SLLx$.

Problem 10

Suppose the combinators B, W and S. Recall that, because BWB is a lark, it follows by Problem 9 that $S(BWB)(BWB)$ is a sage. In turn, $WS(BWB)$ is a sage, as $WS(BWB) = S(BWB)(BWB)$.

Note: Smullyan says that $S(BWB(BWB))$ is a sage, but this seems incorrect. We know that SLL is a sage, not $S(LL)$.

Problem 11

Suppose the combinators M and P are present. Take the sage θ to be $PM(PM)$ (that is, $M(PM)$), as $PM(PM)x = x(M(PM)x) = x(PM(PM)x)$.

Problem 12

Suppose the combinators P and W are present. Take the sage θ to be $WP(WP)$, as $WP(WP)x = P(WP)(WP)x = x(WP(WP)x)$.

Problem 13

Suppose the combinators P and I are present. Recall that PII is a mocker, as $PIIx = x(IIx) = x(Ix) = x(x) = x$. Thus, by Problem 11, $P(PII)(P(PII))$ is a sage (or, more simply, $PII(P(PII))$).

Problem 14

Take U to be W_1P_1, as $W_1P_1xy = P_1xyy = y(xxy)$.

Problem 15

Take U to be WP, as $WPxy = Pxxy = y(xxy)$.

Problem 16

Take U to be PM, as $PMxy = y(Mxy) = y(xxy)$.

Problem 17

Take U to be $L(SI)$, as $L(SI)xy = SI(xx)y = Iy((xx)y) = y(xxy)$.

Problem 18

Take θ to be UU, as $UUx = x(UUx)$.

Problem 19

Take O to be QQW, as $QQWxy = W(Qx)y = Qxyy = y(xy)$.

Problem 20

Take O to be $BW(CB)$, as $BW(CB)xy = W(CBx)y = CBxyy = Byxy = y(xy)$.

Problem 21

Take O to be SI, as $SIxy = Iy(xy) = y(xy)$.

Problem 22

Take O to be PI, as $PIxy = y(Ixy) = y(xy)$.

Problem 23

Recall that, if B and M are present, $M(BxM)$ is a fixed point for every x. Thus $M(BOM)$ is a fixed point of O. That is a sage, as $M(BOM)x = O(M(BOM))x = x(M(BOM)x)$.

Problem 24

Recall that, if L is present, $Lx(Lx)$ is a fixed point of every x. Thus $LO(LO)$ is a fixed point of O. That is a sage, as $LO(LO)x = O(LO(LO))x = x(LO(LO)x)$.

Problem 25

Take U to be BOM, as $BOMxy = O(Mx)y = y(Mxy) = y(xxy)$.

Problem 26

Take U to be LO, as $LOxy = O(xx)y = y(xxy)$.

Problem 27

Suppose O' is a fixed point of O. Thus $O'x = OO'x = x(O'x)$, meaning O' is a sage.

Problem 28

Suppose θ is a sage. Thus $\theta x = x(\theta x)$, and $O\theta x = x(\theta x) = \theta x$, meaning θx is a fixed point of O for every x. Smullyan seems to assume that we could take x to be an empty element, which would of course mean that $O\theta = \theta$, that is, θ is a fixed point of O.

Problem 29

Suppose we have a nice combinator N. Now consider any pair of elements (x_1, x_2). Let $y_1 = Nx_1x_1x_2$, and $y_2 = Nx_2x_1x_2$. Then $y_1 = Nx_1x_1x_2 = x_1(Nx_1x_1x_2)(Nx_2x_1x_2)$ and $y_2 = Nx_2x_1x_2 = x_2(Nx_1x_1x_1)(Nx_2x_1x_2)$, showing (y_1, y_2) is a double fixed point of (x_1, x_2), meaning the weak double fixed point property holds.

Problem 30

(a): Take n to be $C^{**}V^*\Phi_3W(C(W^{**}R^*))$, as

$$C^{**}V^*\Phi_3W(C(W^{**}R^*))wzxy = V^*\Phi_3W(C(W^{**}R^*))zwxy$$

$$= \Phi_3zW(C(W^{**}R^*))wxy = z(Wwxy)(C(W^{**}R^*)wxy)$$

$$= z(wxxy)(W^{**}R^*xwy) = z(wxxy)(R^*xwyy) = z(wxxy)(wyxy)$$

Alternatively, we could, as Smullyan does, take n to be $C(V^*\Phi_3W(W_2C))$, as

$$C(V^*\Phi_3W(W_2C))wzxy = V^*\Phi_3W(W_2C)zwxy$$

$$= \Phi_3zW(W_2C)wxy = z((Wwxy)(W_2Cwxy) = z(wxxy)(Cwxyy) = z(wxxy)(wyxy)$$

(b): Recall by Problem 35 of Chapter 9 that $L = B(TM)B$, meaning a lark is present, and that $M(Lx)$ is a fixed point for every x. Thus $M(Ln)$ is a fixed point of n. Now suppose that n' is any fixed point of n, that is, $nn' = n'$. Thus $n'zxy = nn'zxy = z(n'xxy)(n'yxy)$.

Problem 31

Take θ_1 to be WN, and θ_2 to be $W_1(CN)$. Note that $WNxy = Nxxy$, and $W_1(CN)xy = CNxyy = Nyxy$. Thus

$$\theta_1xy = WNxy = Nxxy = x(Nxxy)(Nyxy) = x(\theta_1xy)(\theta_2xy)$$

Similarly,

$$\theta_2xy = W_1(CN)xy = CNxyy = Nyxy = y(Nxxy)(Nyxy) = y(\theta_1xy)(\theta_2xy)$$

meaning (θ_1, θ_2) is a double sage.

Problem 32

First, we can derive O_1 and O_2 from B, T and M. Take O_1 to be $W_2(V^*\Phi_2)$ and O_2 to be $W_3(C^{***}V^*\Phi_2)$, as

$$O_1zwxy = W_2(V^*\Phi_2)zwxy = V^*\Phi_2zwxxy = \Phi_2xzwxy = x(zxy)(wxy)$$

and

$$O_2zwxy = W_3(C^{***}V^*\Phi_2)zwxy = C^{***}V^*\Phi_2zwxyy = V^*\Phi_2zwyxy = \Phi_2yzwxy = y(zxy)(wxy)$$

Now, consider any O_1, O_2. Now consider some double fixed point (y_1, y_2) of (O_1, O_2). Thus $O_1y_1y_2 = y_1$ and $O_2y_1y_2 = y_2$. Thus, for every x and y,

$$y_1xy = O_1y_1y_2xy = x(y_1xy)(y_2xy)$$

and

$$y_2xy = O_2y_1y_2xy = y(y_1xy)(y_2xy)$$

meaning (y_1, y_2) is a double sage.

We can verify the statement that every double sage is a double fixed point of any dobule oracle in the following way. In particular, we show this for the double oracle (O_1, O_2) (this is what Smullyan did, and I could not find a result for double oracles in general). Now consider any double sage (θ_1, θ_2). Thus

$$\theta_1xy = x(\theta_1xy)(\theta_2xy)$$

$$\theta_2xy = y(\theta_1xy)(\theta_2xy)$$

for all x and y. But also

$$O_1\theta_1\theta_2xy = x(\theta_1xy)(\theta_2xy)$$

$$O_2\theta_1\theta_2xy = y(\theta_1xy)(\theta_2xy)$$

for all x and y. Thus

$$O_1\theta_1\theta_2xy = \theta_1xy$$

and

$$O_2\theta_1\theta_2xy = \theta_2xy$$

Taking x and y for empty elements, we receive $O_1\theta_1\theta_2 = \theta_1$ and $O_2\theta_1\theta_2 = \theta_2$, meaning (θ_1, θ_2) is a double fixed point of (O_1, O_2).